

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 3.4: Magnetism and Electromagnetic Induction — Teacher Reference (Pre-filled)

ATTRIBUTION AND LICENSING

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SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand	Sub-Strand 3.4: Magnetism and Electromagnetic Induction
Driving Question	DRIVING QUESTION: How does moving a wire through a magnetic field produce electricity — and how is this principle powering homes, farms and industries across Kenya?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Methods of Magnetisation	Students observe that a steel nail rubbed repeatedly in one direction with a bar magnet picks up paper clips, while an un-stroked nail does not. Teacher should cold-call at least 3 students to describe exactly what they see before explaining why. Wait time: 30 seconds after each prompt. Students also observe (via video or demonstration) a large generator at Olkaria Geothermal Plant spinning and lighting a bulb — the anchoring phenomenon for the entire sub-strand.	Soft iron and steel can be magnetised by three methods: (1) STROKING — repeated unidirectional strokes with one pole of a bar magnet align magnetic domains in one direction; (2) ELECTRICAL METHOD — passing direct current through a solenoid wrapped around the material aligns domains using the magnetic field of the coil; (3) HAMMERING IN A MAGNETIC FIELD — mechanical energy while the material is held along Earth's magnetic field lines helps domains settle in alignment. Pedagogical	The nail becomes a magnet because its previously random magnetic domains (tiny regions of aligned atomic magnets) are forced into a common orientation by the external magnetic field of the stroking magnet. Steel retains this alignment (hard magnetic material) while soft iron loses it quickly. Teacher should link back to the anchoring phenomenon: the rotating coils inside the Olkaria generator are moving through a magnetic field — the sub-strand will explain how that motion generates the

		note: emphasise that all three methods share the same underlying mechanism — domain alignment — even though the energy source differs.	electricity that lights Kenyan homes.
Lesson 2 Demagnetisation & Properties of Magnets: Why Stillness Kills the Current	Students observe a magnetised nail before and after it is (a) heated over a Bunsen burner, (b) dropped repeatedly on the bench, and (c) placed near an alternating-current coil. In each case they test the nail's ability to pick up paper clips before and after treatment. Teacher cold-calls 4 students — two to predict outcomes before each treatment, two to report after. Wait time: 20 seconds. Students record observations in a results table.	Magnetism depends on the ordered alignment of magnetic domains. Any process that supplies sufficient energy disrupts this alignment and destroys magnetism. The three demagnetisation methods are: (1) HEATING beyond the Curie point — thermal energy randomises domain orientation; (2) HAMMERING/MECHANICAL SHOCK — vibration randomly re-orientates domains; (3) ALTERNATING CURRENT (A.C.) — rapidly reversing field repeatedly realigns then randomises domains until net magnetisation is zero. Key properties of magnets reviewed: attraction and repulsion (like poles repel, unlike attract), directive property (freely suspended magnet aligns N–S), and magnetic induction (magnets can induce temporary magnetism in nearby materials). Pedagogical note: the title 'Why Stillness Kills the Current' foreshadows Lesson 4 — a stationary magnet near a wire induces NO e.m.f.	Demagnetisation is the reverse of magnetisation: instead of aligning domains, energy is added to randomise them. Teachers should connect this explicitly to the driving question — when the Olkaria turbine STOPS spinning, the relative motion between coil and magnetic field ceases and current drops to zero, just as a heated magnet loses its ordered domains. Both phenomena are about order versus disorder at the microscopic level.
Lesson 3 Magnetic Field Patterns: Mapping the Invisible Force	Students place a bar magnet under a sheet of white paper and sprinkle iron filings evenly over the surface. They gently tap the paper and observe the filings aligning into curved patterns. Teacher directs students to sketch the patterns and label North and South poles, field line direction, and regions of high vs. low field density. Cold-call 3 students to describe a specific feature (e.g., spacing near the poles vs. far away). Wait time: 25 seconds. A second activity uses a plotting compass to trace field lines around unlike-pole and like-pole configurations.	Magnetic fields have direction: field lines run from the North pole to the South pole OUTSIDE the magnet, and from South to North INSIDE. Field line density (closeness of lines) indicates field strength — lines are closest near the poles where the field is strongest and spread out further away where it is weakest. Key patterns: (a) single bar magnet — oval loops converging at poles; (b) unlike poles facing — lines curve from N to S, creating attraction; (c) like poles facing — lines repel, creating a neutral point between the magnets where field strength is zero. Pedagogical note: the concept of field line density directly prepares students for Lesson 5, where greater field strength produces greater induced e.m.f.	A magnetic field is a region of space where a magnetic force is experienced. Field lines are a visual model — they do not physically exist but represent the direction a free North pole would travel. Teacher should ask: 'If we could shrink and ride a field line from the North pole of a magnet at the Rift Valley, where would we end up?' (South pole — outside the magnet.) Connecting to the driving question: the strong magnetic fields inside generators at Gitaru Hydroelectric Dam are what make electromagnetic induction powerful enough to supply electricity to the national grid.

Lesson 4 Introducing Induced E.M.F (Qualitative): What Makes the Galvanometer Deflect?	Students connect a coil of wire to a sensitive galvanometer (centre-zero). They systematically: (a) hold a bar magnet stationary inside the coil — observe zero deflection; (b) push the magnet INTO the coil — observe deflection in one direction; (c) hold the magnet still again — observe zero deflection; (d) pull the magnet OUT — observe deflection in the opposite direction; (e) push faster — observe larger deflection; (f) use a stronger magnet — observe larger deflection. Teacher cold-calls 5 students at different steps to state exactly what the galvanometer needle does. Wait time: 30 seconds after each observation step. Students record all results in a systematic table.	An electromotive force (e.m.f.) is induced in a conductor ONLY when there is relative motion between the conductor and the magnetic field. Key qualitative rules: (1) No motion = no induced e.m.f. (galvanometer reads zero); (2) Motion into the field = deflection in one direction (conventional current flows one way); (3) Motion out of the field = deflection in the opposite direction (current reverses); (4) Greater speed or stronger magnet = greater deflection (larger induced e.m.f.). This is Faraday's Law of Electromagnetic Induction in qualitative form. Pedagogical note: the title 'Why Stillness Kills the Current' from Lesson 2 pays off here — a stationary magnet, like a demagnetised nail, produces no useful effect.	When the magnet moves into the coil, it changes the number of magnetic field lines (magnetic flux) passing through the coil. This changing flux forces electrons in the wire to move — creating an induced current. When flux is constant (magnet still), electrons have no 'push' and current is zero. Teacher analogy for Kenyan context: 'Think of ugali being stirred — when the wooden spoon moves, it drags the ugali with it. When the spoon is still, the ugali stays still too. The spoon is the magnet; the ugali is the electrons.' Connect to driving question: Olkaria's turbines spin continuously — continuous motion = continuous changing flux = continuous electricity supply to the national grid.
Lesson 5 Factors Affecting the Magnitude of Induced E.M.F.	Students conduct a structured investigation varying ONE factor at a time while keeping others constant, recording galvanometer deflection (as a proxy for e.m.f.) for each: (1) SPEED — push magnet slowly vs. quickly; (2) FIELD STRENGTH — use a weak magnet vs. a strong magnet; (3) NUMBER OF TURNS — coil with 20 turns vs. 50 turns vs. 100 turns; (4) CROSS-SECTIONAL AREA OF COIL — narrow coil vs. wide coil with same number of turns. Teacher cold-calls 2 students per variable to state the independent variable, dependent variable, and one controlled variable. Wait time: 20 seconds. Students plot bar graphs for each variable.	Four factors control the magnitude of induced e.m.f.: (1) SPEED OF RELATIVE MOTION — faster motion = greater rate of change of flux = larger e.m.f.; (2) STRENGTH OF MAGNETIC FIELD — stronger field = more flux lines cut per second = larger e.m.f.; (3) NUMBER OF TURNS IN THE COIL — more turns = more conductors cutting flux simultaneously = larger e.m.f. (this is why generator coils have many hundreds of turns); (4) CROSS-SECTIONAL AREA OF THE COIL — larger area = more flux passes through = larger e.m.f. when flux changes. Pedagogical note: each factor maps directly to a design choice engineers made when building Kenya's power stations — large coils, many turns, strong electromagnets, and turbines designed to spin at optimal speed.	This lesson quantifies what Lesson 4 showed qualitatively. Teacher should ask students to rank which factor would be most practical to increase in a real power station (number of turns and field strength are most cost-effective to engineer). Connect to driving question: engineers at the Lake Turkana Wind Farm in Marsabit County designed the generator coils with hundreds of turns and powerful electromagnets precisely because increasing these factors multiplies the e.m.f. — the same physics, scaled up to produce 310 MW of power for Kenya's national grid.
Lesson 6 Applications of Electromagnetic Induction in Kenya	Students examine (via physical samples, diagrams, or video clips) six devices that exploit electromagnetic induction: (1) Bicycle dynamo — students see the rotating magnet near a coil; (2) A.C. generator — cutaway diagram showing slip rings and rotating coil; (3) D.C. generator — cutaway showing split-ring commutator; (4) Transformer (step-up and step-down) —	All six applications operate on the same principle: relative motion (or changing current) between a conductor and a magnetic field induces an e.m.f. Specific Kenyan connections: (a) BICYCLE DYNAMO — used by students in Kericho and rural areas to power bicycle lights; the spinning wheel rotates a magnet past a coil; (b) A.C. GENERATOR (alternator) — used	Teacher should draw a concept map on the board connecting all six applications back to Faraday's Law. Ask: 'Which of these six devices is most relevant to YOUR home or community?' Allow students 2 minutes of pair discussion, then cold-call 4 pairs. The answer to the driving question is now largely complete: moving a wire (or magnet) through a magnetic field changes magnetic

	diagram of iron core with primary and secondary coils; (5) Induction cooker — diagram showing coil beneath the cooking surface; (6) Lake Turkana Wind Farm — image/video of turbines. Teacher cold-calls 3 students to identify the 'moving part' or 'changing current' that drives induction in each device. Wait time: 25 seconds per prompt.	at Gitaru Hydroelectric Dam; slip rings allow the coil to rotate continuously, producing alternating current; (c) D.C. GENERATOR — split-ring commutator rectifies the current so it flows in one direction only; (d) TRANSFORMER — steps voltage up for long-distance transmission across Kenya (e.g., Nairobi to Mombasa high-voltage lines) and steps it down for safe home use; (e) INDUCTION COOKER — alternating current in a coil creates a changing magnetic field that induces currents directly in the cooking pot, generating heat — no open flame, important for safety in urban Nairobi homes; (f) WIND TURBINE — at Lake Turkana, wind rotates large blades connected to a generator shaft. Pedagogical note: transformer operation relies on CHANGING flux (from A.C.), not mechanical motion — an important conceptual extension.	flux, which induces an e.m.f., which drives a current — and this principle, scaled up at Olkaria, Gitaru, Lake Turkana, and Masinga Dam, is literally powering Kenya's homes, farms (irrigation pumps), and industries (Nairobi factories).
Lesson 7 Project: Design and Make a Simple Electric Bell	Student teams build an electric bell using: an iron nail wrapped with insulated copper wire (electromagnet), a springy steel strip (armature), a metal contact screw, a battery, and a bell/striker. They observe: when the circuit is closed, the nail attracts the armature, the striker hits the bell, the circuit breaks, the nail demagnetises, the spring pulls the armature back, the circuit closes again — producing a rapid, continuous ringing cycle. Teacher directs teams to count the number of attract-release cycles per second by listening and timing. Cold-call 3 teams to explain one specific step in the cycle. Wait time: 30 seconds.	The electric bell works because of controlled, repeated magnetisation and demagnetisation of soft iron. The iron nail becomes an electromagnet (magnetised) when current flows, attracting the soft-iron armature. When the armature moves, it breaks the circuit, current stops, the nail demagnetises (soft iron loses magnetism quickly — revisiting Lesson 1), and the spring returns the armature to restart the cycle. Soft iron is deliberately chosen because it magnetises and demagnetises rapidly and repeatedly. Pedagogical note: this project consolidates Lessons 1 (magnetisation by electrical method), 2 (demagnetisation/properties), and 4–5 (the importance of CHANGING conditions — here, switching current on and off — to produce dynamic magnetic effects). Students apply the full engineering design process: plan, build, test, improve.	The electric bell is a physical model of the core principle of the sub-strand: CHANGE is required for magnetic effects to do useful work. A constant current would hold the armature permanently — no ringing. An interrupted current produces repeated magnetisation/demagnetisation cycles — useful mechanical work (sound). Teacher should ask: 'How is the on-off switching of current in the bell similar to the spinning of a coil in a generator?' (Both involve continuously changing magnetic conditions.) Final connection to the driving question: engineering solutions — from the school electric bell to the Olkaria turbine — all exploit the dynamic relationship between electricity and magnetism that Faraday discovered.
Lesson 8 Assessment & Driving Question Resolution: Powering Kenya	Students revisit the anchoring phenomenon from Lesson 1 (the Olkaria Geothermal Plant generator video). Teacher replays the clip and asks students to annotate a	Students consolidate the full conceptual arc of the sub-strand: (1) Magnetisation aligns magnetic domains (Lessons 1–2); (2) Magnetic fields have direction and strength	The driving question is fully resolved: moving a wire through a magnetic field changes magnetic flux, inducing an e.m.f. by Faraday's Law. This principle,

Through Electromagnetic Induction	generator diagram identifying: the coil, the magnetic field, the direction of motion, the slip rings, and the output terminals. Students also review their Driving Question Board entries from all seven lessons and identify which questions have been answered and which remain open. Cold-call 5 students to read one DQB entry and explain how the sub-strand answered it. Wait time: 20 seconds per student.	represented by field lines (Lesson 3); (3) Relative motion between a conductor and magnetic field induces e.m.f. — Faraday's Law (Lesson 4); (4) Greater speed, field strength, number of turns, and coil area increase the induced e.m.f. (Lesson 5); (5) These principles are applied in dynamos, generators, transformers, and wind turbines across Kenya (Lesson 6); (6) Electromagnetic switching underpins devices like the electric bell (Lesson 7). Pedagogical note: the assessment should include both knowledge recall AND application to novel Kenyan contexts (e.g., 'A farmer in Nakuru wants to use a small river to generate electricity for irrigation pumps — what factors should the generator designer consider?').	discovered in a London laboratory in 1831, is today harnessed at Olkaria (geothermal, 861 MW), Gitaru (hydroelectric, 225 MW), Lake Turkana (wind, 310 MW), and Masinga Dam (hydroelectric, 40 MW) to power homes, run irrigation pumps on Kenyan farms, and drive motors in Nairobi's industrial area. Teacher should close by asking: 'What would Kenya look like without electromagnetic induction?' — allowing students to appreciate the profound real-world significance of the physics they have studied.
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